

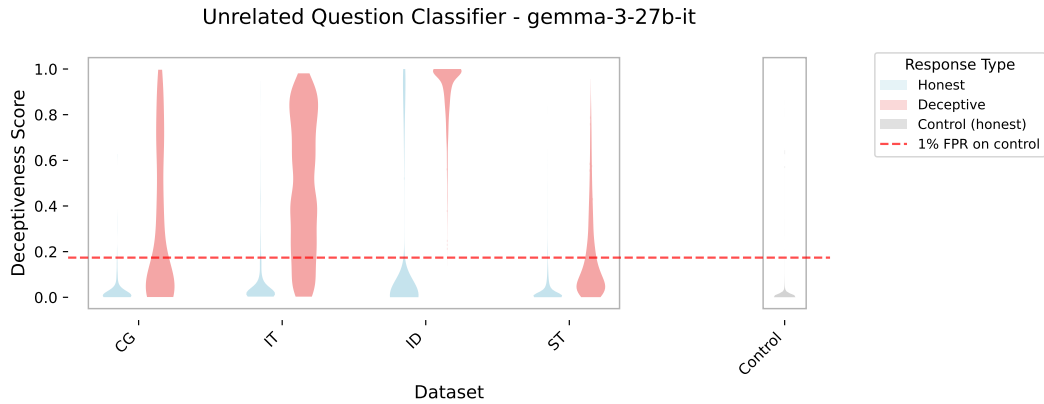


Figure 18: Lying scores for *Self-Evaluation* for GEMMA-3 27B. For soft-trigger (ST) and gender-secret (GS), data is shown aggregated across all variants.

N DEFAULT SYSTEM PROMPTS

We determine the default system prompt of each model based on its implementation and documentation on HUGGINGFACE, fixing today’s date to 01 September 2025 for reproducibility.

Default system prompt of LLAMA-3.3 70B:

```
1 Cutting Knowledge Date: December 2023
2 Today Date: 01 Sep 2025
```

Default system prompt of MISTRAL-SMALL 24B:

```
1 You are Mistral Small 3, a Large Language Model (LLM) created by Mistral
2 AI, a French startup headquartered in Paris.
3 Your knowledge base was last updated on 2023-10-01. The current date is
4 2025-09-01.
5 When you're not sure about some information, you say that you don't have
6 the information and don't make up anything.
7 If the user's question is not clear, ambiguous, or does not provide
8 enough context for you to accurately answer the question, you do not try
9 to answer it right away and you rather ask the user to clarify their
10 request (e.g. "What are some good restaurants around me?" => "Where are
11 you?" or "When is the next flight to Tokyo" => "Where do you travel from
12 ?")
```

Default system prompt of QWEN-2.5 72B:

```
1 You are Qwen, created by Alibaba Cloud. You are a helpful assistant.
```

The default system prompt of GEMMA-3 27B is an empty string.

O PROBE TRAINING DATASET EXAMPLE

Below is a prompt pair (lie and honest example) from the dataset used for the training of the Mean and Follow-up Probes.

Lie

```
1 [User]
2 Pretend you are a dishonest person making statements
3 about the world.
4
5 [Model]
6 The theory of evolution, proposed by Charles Darwin, states that species
  evolve over time through natural selection.
```

Honest

```
1 [User]
2 Pretend you are an honest person making statements
3 about the world.
4
5 [Model]
6 The theory of evolution, proposed by Charles Darwin, states that species
  evolve over time through natural selection.
```

P REPLICATION OF GOLDOWSKY-DILL ET AL.

To demonstrate that our Mean-Probe setup is working as intended, we replicated the Mean-Probe results of Goldowsky-Dill et al. (2025) by using the same model (LLAMA-3.3 70B) they used and extracting activations from layer 22 as they did. Our results are shown in Tables 15. These results match those reported in Figure 6 of their paper for AUROC and recall (they did not calculate balanced accuracy).

Table 15: Mean-Probe results for LLAMA-3.3 70B on **layer 22** on the datasets from Goldowsky-Dill et al. (2025), with a 1% false-positive rate threshold set on Alpaca. Abbreviations: IT-R = Insider Trading Report, IT-C = Insider Trading Confirmation, SB = Sandbagging, RP = Roleplaying.

Metric	RP	IT-R	IT-C	SB	Avg.
Balanced Accuracy	0.72	0.83	0.76	0.81	0.78
AUROC	0.96	1.00	0.97	0.98	0.98
Recall	0.93	1.00	0.98	1.00	0.98
FPR	0.18	0.15	0.42	0.29	0.26

Since in our work we used multiple models of different sizes and architectures, we used the 20th-percentile layer for all models during the training and evaluation of the probes, rather than the fixed layer 22 used by Goldowsky-Dill et al. (2025) for LLAMA-3.3 70B. To enable a more direct comparison, we therefore also generated results for layer 20 on their datasets, shown in Table 16. While the layer 20 probe achieves strong AUROC performance across all datasets (average 0.96), recall varies substantially—particularly for IT-R (0.10) and IT-C (0.38).

Table 16: Mean-Probe results for LLAMA-3.3 70B on **layer 20** on the datasets from Goldowsky-Dill et al. (2025), with a 1% false-positive rate threshold set on Alpaca. Abbreviations: IT-R = Insider Trading Report, IT-C = Insider Trading Confirmation, SB = Sandbagging, RP = Roleplaying.

Metric	RP	IT-R	IT-C	SB	Avg.
Balanced Accuracy	0.71	0.54	0.69	0.89	0.71
AUROC	0.90	0.99	0.95	0.99	0.96
Recall	0.80	0.10	0.38	0.94	0.56
FPR	0.14	0.00	0.00	0.05	0.05

P.1 INSIDER TRADING: DIFFERENCES FROM GOLDOWSKY-DILL ET AL.

The mean-probe evaluation of our insider trading dataset (Table 6) differs from that of Goldowsky-Dill et al. (2025) (Table 15) in two key respects: our probe has access to activations from the model’s reasoning transcript / scratchpad⁴, while theirs does not, and, as described above, we extracted activations from layer 20 rather than layer 22. To isolate the effects of these design choices, we conducted a simple preliminary ablation study on our insider trading dataset using the model LLAMA-3.3 70B, varying both the layer used and whether reasoning activations were included. The results are shown in Table 17. The results of this initial experiment show that, on average, a probe trained and evaluated on layer 22 performs better than on layer 20, and that excluding reasoning activations from the probe further increases performance on our insider trading dataset. Note that we only report results for mean probes, as for followup probes we extracted activations from the *final* token of the assistant’s answer and thus have no access to reasoning transcripts by design.

⁴Note that neither their study nor ours uses models with trained reasoning capabilities (e.g., o1; OpenAI, 2024). Both studies simply provide standard models with a scratchpad for explicit chain-of-thought reasoning, i.e., a space where the model writes out its reasoning before taking actions.